Kaneda Fabrication Works (fictional)

WELDING PROCEDURE SPECIFICATION (WPS)

Format per ASME BPVC Section IX, Form QW-482 (see QW-200.1)

WPS no.	**WPS-09**
Revision	**0**
Date	**2024-05-20**
Supporting PQR	**PQR-09**

Welding process(es)	Type(s)
FCAW	**Semi-automatic**

JOINTS (QW-402)

Joint design	Backing
Fillet welds, all sizes; single-bevel groove 45°, root gap 0-3 mm	**Optional (steel)**

BASE METALS (QW-403)

P-No.	Group No.	to P-No.	Group No.
1	**1 or 2**	**1**	**1 or 2**

Thickness range, base metal - groove	Fillet
1.5 to 20 mm	**All**

FILLER METALS (QW-404)

Process	Spec. No. (SFA)	AWS No. (Class)	F-No.	A-No.	Size	Deposited thickness
FCAW	SFA-5.20	E71T-1C	6	1	1.2 mm	max. 20 mm

POSITIONS (QW-405)

Position(s) of groove	Position(s) of fillet	Welding progression
All (groove and fillet)	**All**	**Uphill**

PREHEAT (QW-406)

Preheat temperature, minimum	Interpass temperature, maximum
10 °C minimum (to 25 mm)	**250 °C maximum**

POSTWELD HEAT TREATMENT (QW-407)

PWHT	Temperature range	Time range
None	**None**	**None**

GAS (QW-408)

Shielding: CO2 100 %, 15-20 L/min.

ELECTRICAL CHARACTERISTICS (QW-409)

Weld pass(es)	Process	Classification	Diameter (mm)	Current / polarity	Amps (range)	Volts (range)	Travel (mm/min)
All	FCAW	E71T-1C	1.2	DCEP	180 - 260	24 - 29	250 - 400

TECHNIQUE (QW-410)

String or weave bead: stringer, weave max. 3 x electrode diameter. Multiple passes per side.

Initial and interpass cleaning: brushing and grinding. Back gouging: grinding to sound metal.

Prepared by (welding engineer)	Approved by (QA manager)
K. Mori (fictional)	**L. Ortega (fictional)**
signed K. Mori 2024-05-20	*signed L. Ortega* 2024-05-20

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Kaneda Fabrication Works (fictional)

PROCEDURE QUALIFICATION RECORD (PQR)

Format per ASME BPVC Section IX, Form QW-483 · actual variables used to weld the test coupon

PQR no.	**PQR-09**
WPS qualified	**WPS-09**
Date welded	**2024-05-02**

TEST COUPON - ACTUAL VARIABLES

Base metals (QW-403)	P-No. to P-No.
SA-36 to SA-36, 10 mm plate	**1 to 1**
Process(es)	**PWHT (QW-407)**
FCAW	**None**

TENSILE TEST (QW-150)

Specimen	Width x thickness (mm)	Ultimate tensile (MPa)	Type of failure and location
T-1	19.0 x 10.0	498	Base metal, ductile
T-2	19.0 x 10.0	505	Base metal, ductile

GUIDED-BEND TESTS (QW-160)

2 face + 2 root bends (QW-462.3(a)), 180°: no open discontinuities. Acceptable.

TOUGHNESS TESTS (QW-170)

Specimen	Test temperature (°C)	Impact values
CVN 10 x 10	-20	Weld 47 / 52 / 49 J

HARDNESS SURVEY (HV10)

Location	Maximum HV10
Weld metal and HAZ, cap and root	236

We certify that the statements in this record are correct and that the test welds were prepared, welded and tested in accordance with the requirements of Section IX of the ASME BPVC.

Mechanical tests by	Certified by (QA manager)
Halvorsen Test Lab (fictional)	**L. Ortega (fictional)**
signed Halvorsen Test Lab — Lab report HTL-09	*signed L. Ortega* — 2024-05-20

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